Embedded Systems Laboratory

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Progress Report

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Outline

- Problem
  - Disk space & internet bandwidth
- Single QP dataset model
  - Flowchart
  - Result & analyze
- Multiple QP dataset model
  - Reason
  - Flowchart
  - Result & analyze
- Future work

Challenges

- **Problems**
  - Limited bandwidth for real-time transmission
  - Insufficient storage capacity
  - Video files too large, hard to manage and analyze

  ⇨ **These issues will become even more critical as the system expands in the future.**

- **Solution**
  - **Video compression** to reduce file size & network bandwidth

Pros and Cons

- **Advantages：**
  - Reduce transmission load
  - Save storage space and bandwidth
  - Improve storage efficiency
- **Disadvantages：**
  - Higher compression ⇨ lower quality ⇨ reduce detection accuracy
  - YOLO is sensitive to details, excessive compression may miss key features
- **Conclusion：**
  - Compression ratio should be maximized within an acceptable accuracy range to achieve the best balance between performance and resource usage

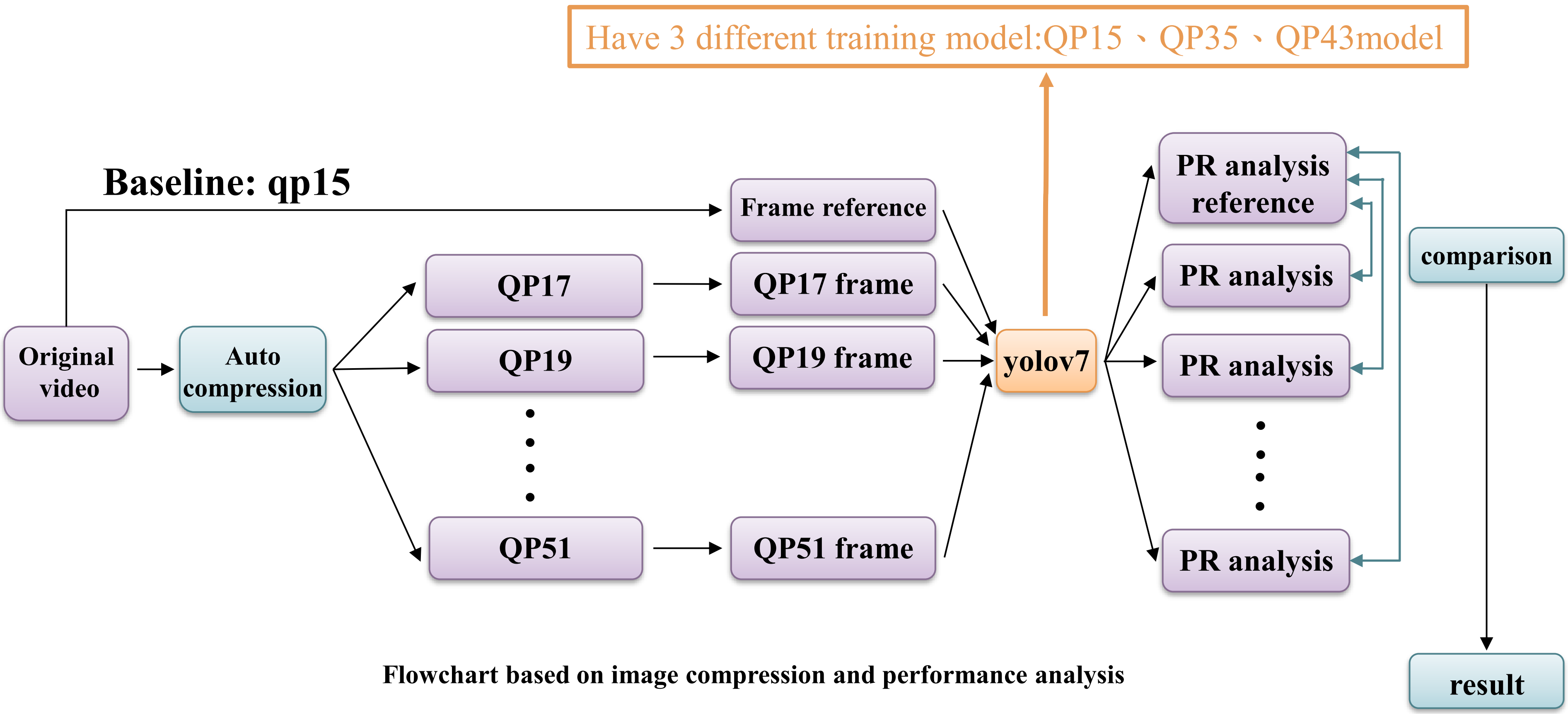
QP(Quantization Parameter)

- QP: a setting that controls how strong the compression is.
  - Higher QP → stronger compression → worse image quality
  - Lower QP → weaker compression → better image quality

Low QP(QP 15)

Higher QP(QP 45)

Method



Flowchart based on image compression and performance analysis

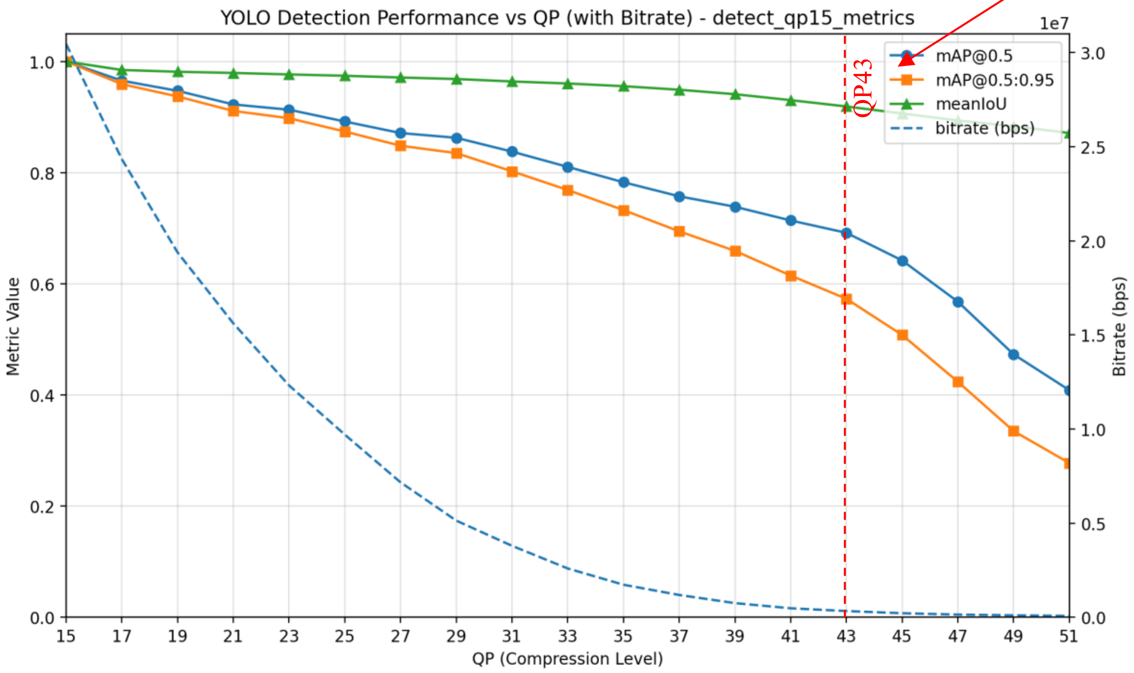
Different Fix QP training model

- 3 different training model: QP15, QP35, and QP43
- Reasons:
  - Model learn from different QP images may affects the detection capability
    - Low QP training model might struggle when detecting the blur images
    - High QP training model may not learn the clear images feature

Result based on qp15-trained model

- **Advantages**: Excellent accuracy on clear, low-QP images.
- **Limitations**: Fails quickly when compression becomes heavy.
- **Conclusion**: Ideal for high-quality streams but performs poorly under heavy compression.

Sharp performance drop beyond QP43



Detect result

QP15

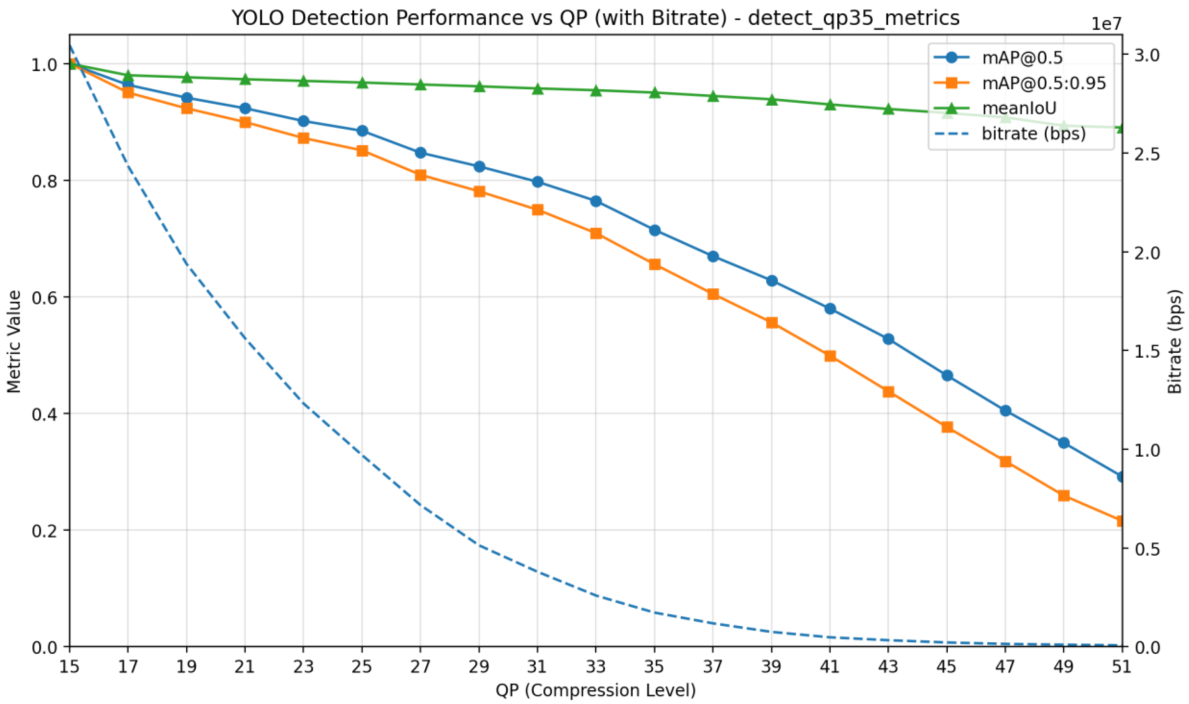
QP41

QP43

QP45

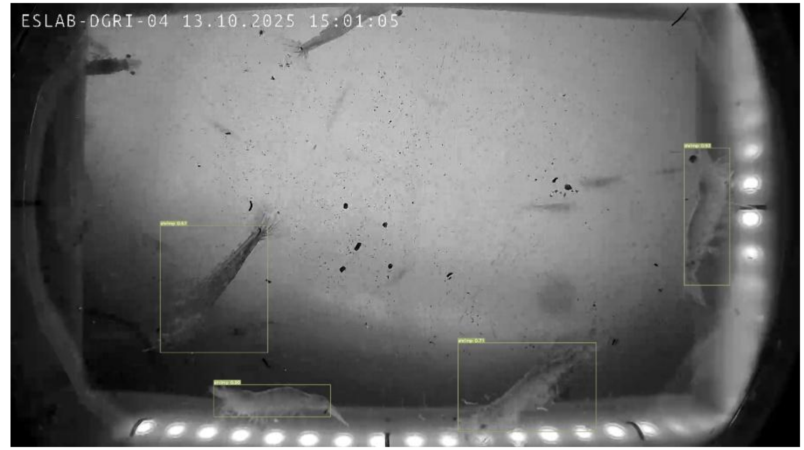
Result based on qp35-trained model

- **Advantages**: More robust to blur and high compression.
- **Limitations**: Weaker on sharp, low-QP images.
- **Conclusion**: Blur-trained models degrade slower under compression than clear-trained models.



Detect result

QP15



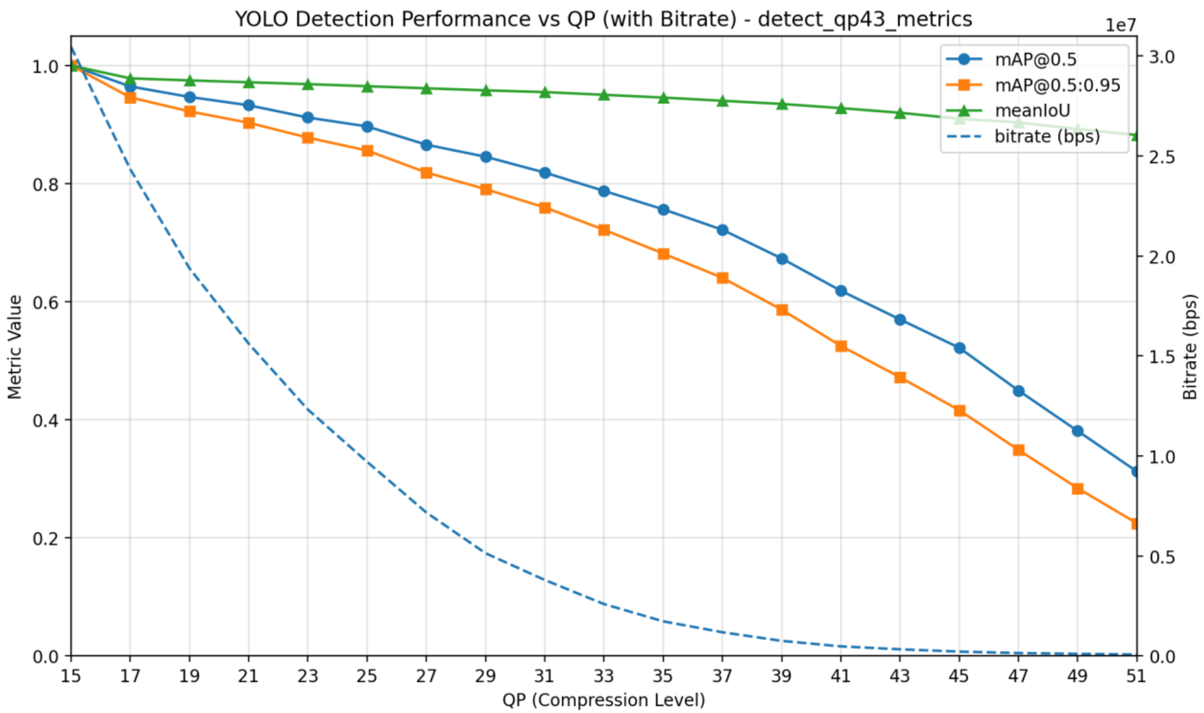
QP41

QP43

QP45

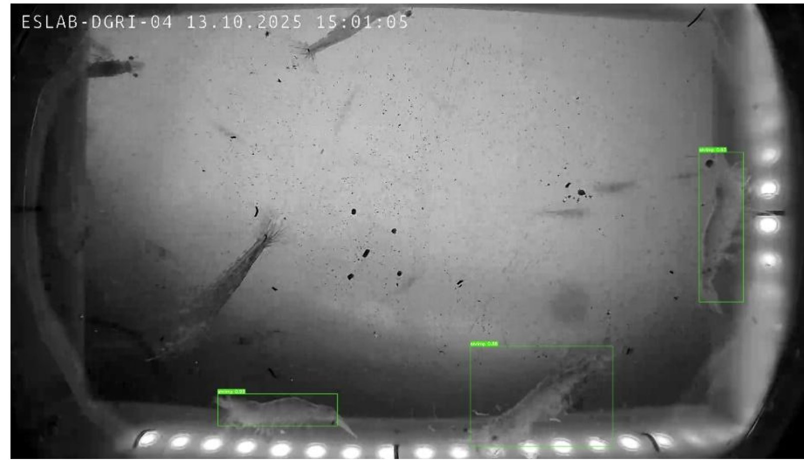
Result based on qp43-trained model

- .**Conclusion:** It seems like QP43 model have similar result with QP35 model



Detect result

QP15



QP41

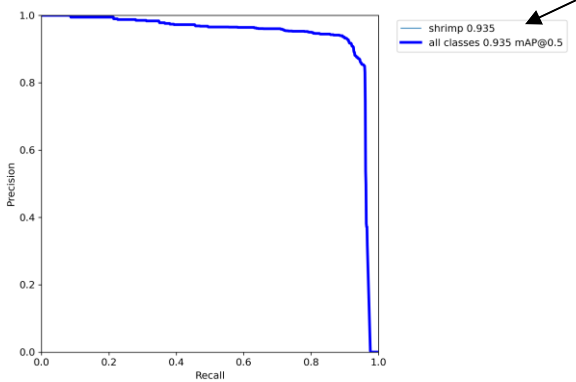
QP43

QP45

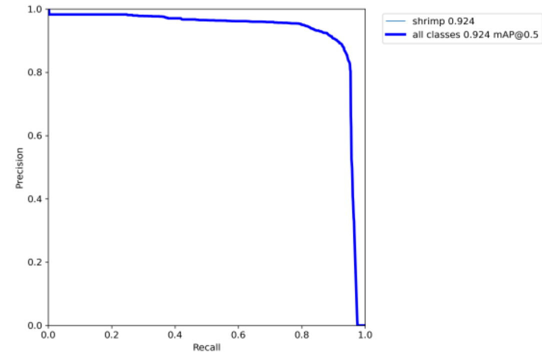
Combine result

The QP15 model achieves the best performance with the highest mAP@0.5 of 0.935.

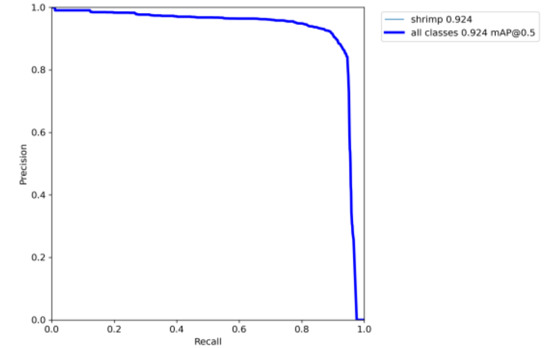
QP15 model



QP35 model

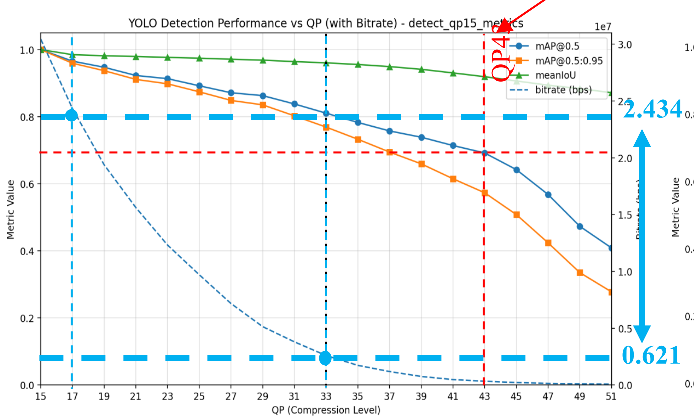


QP43 model

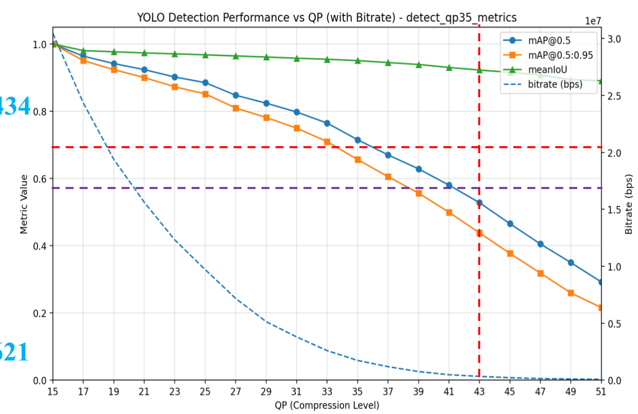


Sharp performance drop beyond QP43

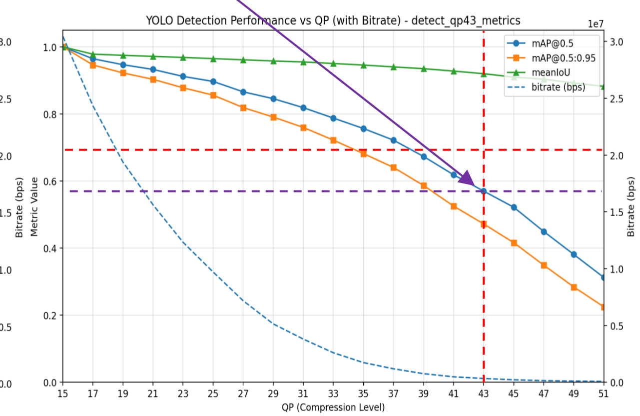
QP43 model remains the more stable under extreme compression compare to QP35 model



Performance under QP15 model



Performance under QP35 model



Performance under QP43 model

Bandwidth requirement is reduced by nearly 4× while maintaining acceptable detection performance.

Extension of the Experiment

- Overall :
  - Although the QP15 model has the best overall performance among all models, QP35 and QP43 remain more stable at higher QP .
- This raises a simple question:
  - Can we train a model that can learn the feature of clear shrimps and the blur shrimps at once?
    - **From Fixed-QP Models to Multiple-QP Learning**

Learning over multiple QP

- Two methods for evaluation:
  - Method1— Mixed-QP Training
    - Each image is randomly assigned a QP or sampled evenly from all QP versions
  - Method2— Curriculum over QP
    - Split different stages **from low QP (easy)** to **high QP (hard)** instead of mixing all QPs at once.

Why Not Simply Do Mixed-QP Training?

- Two perspectives:
  - Early training stages are critical for optimization:
    - High-QP images are extremely blur and noisy
      → early mixing leads to unstable gradients
      → the model ends up stuck in local minima.
  - Feature Learning Order Matters:
    - Low-QP images contain clear and detailed features → these should be learned in the early epochs.
    - Adding high-QP later helps the model stay stable even under compression.

Method 2— Curriculum over QP

- **Three-Stage Training Process**
  - Stage 1 — Low QP Focus (Early Training) **Epoch1~99**

    90% QP15-20 + **10% QP21-25**

  - Stage 2 — Medium Difficulty (Transition Stage) **Epoch100~199**

    60% QP26-35 + **30% QP15-25** + **10% QP36-40**

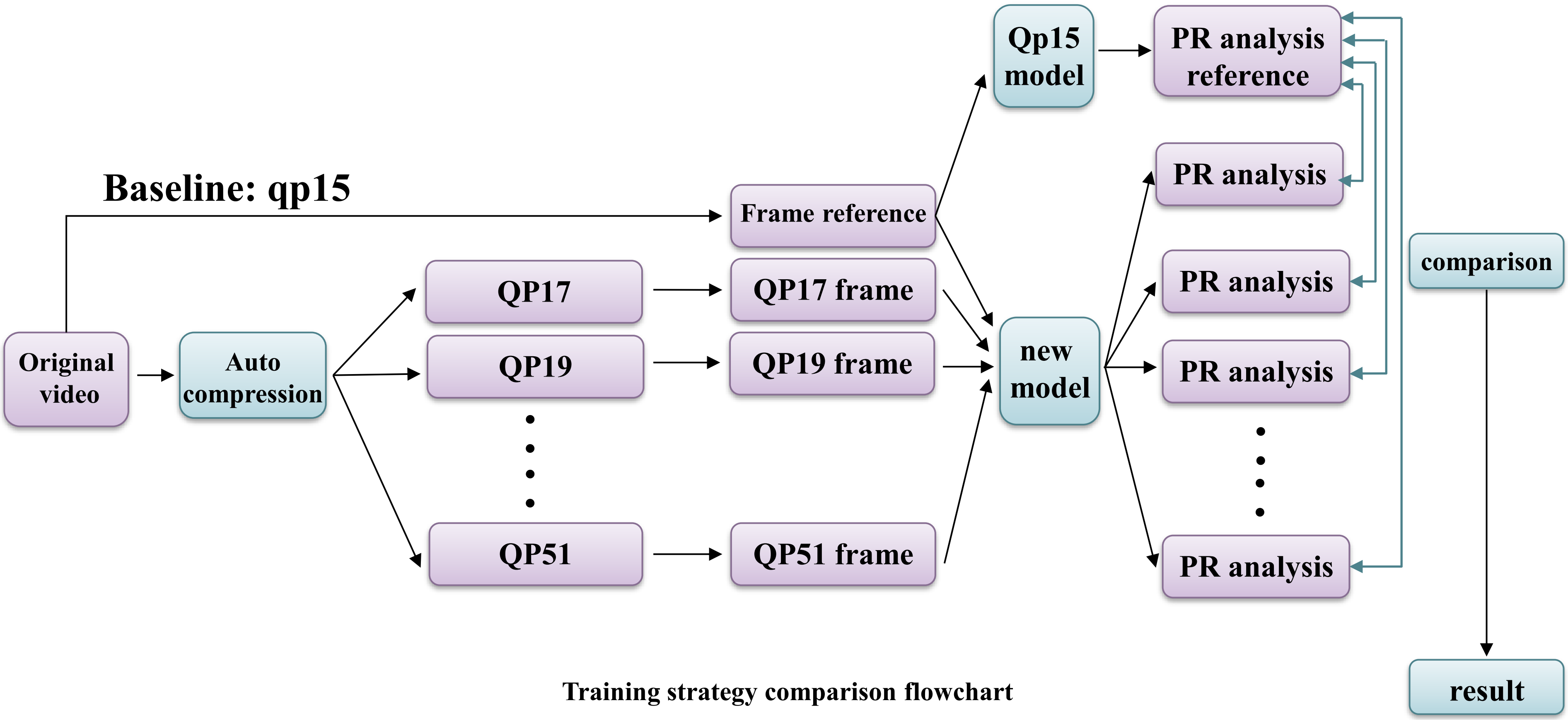
  - Stage 3 — High QP(Late Training) **Epoch200~300**

    60% QP41-45 + **30% QP15-40** + **10% QP46-51**

- **Implementation Details**
  - Use the same number of samples per epoch for all methods.
  - Apply the same learning-rate schedule to every method.

Method

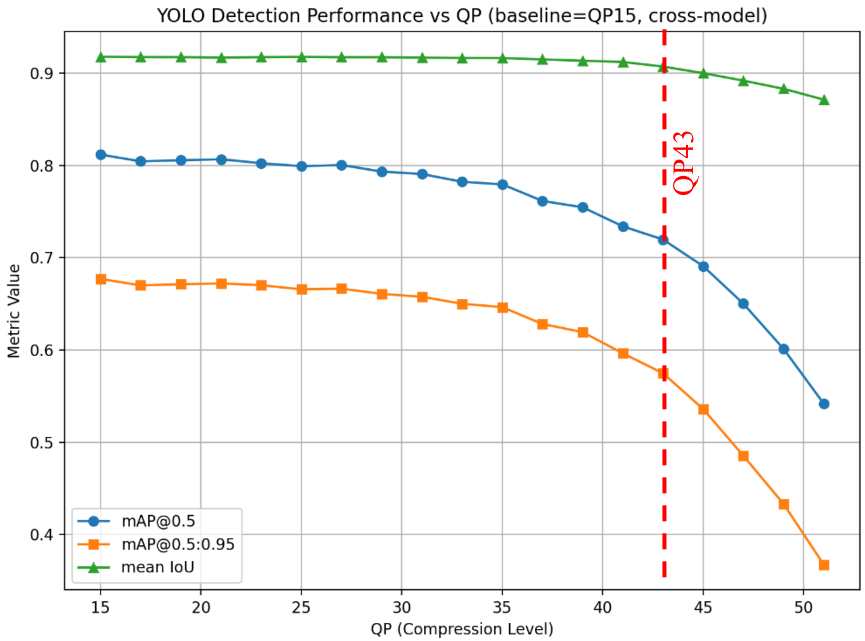
- The goal is to compare different model performance → fix the reference



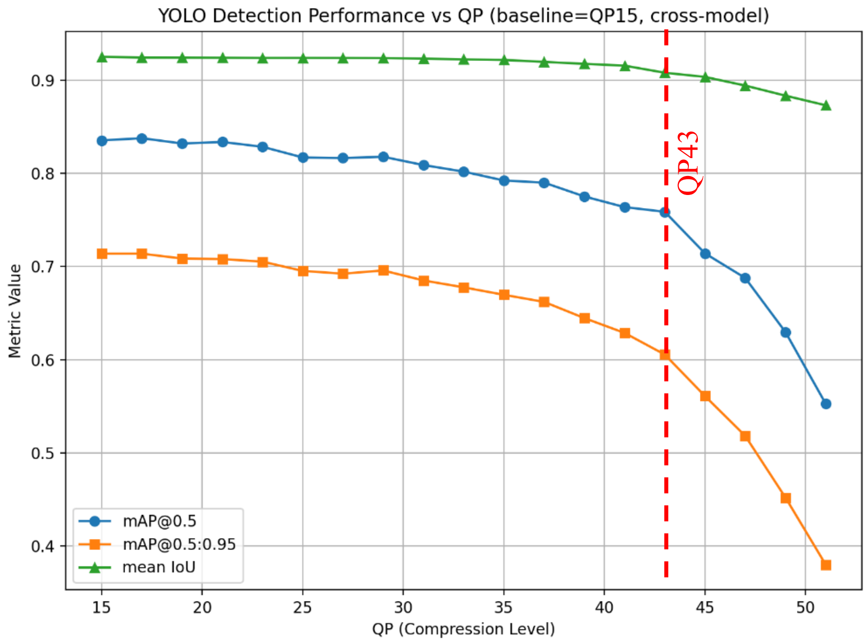
Training strategy comparison flowchart

Compare Result

- Performance improve under Curriculum training: Three metric are consistently higher than Mixed-QP.
- A noticeable decline in detection performance appears after QP43.



Performance under Mix training model



Performance under Curriculum training model